Instructions

There are **50 total points**. This exam is closed-book and closed-notes except for **one 8.5 × 11-inch sheet of paper with hand-written notes (both sides permitted)**. No electronic devices, additional pages, or other materials are permitted. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ✓ ⊙ ⊗ are incorrect . Write your full name in the box to acknowledge the instructions.

Jarrett Chen

Motivation and Applications

1. [3 points]: Since most streaming video became encrypted, which of the following can a network operator *no longer read directly* from the traffic, and so must infer? Select all that apply.

- ● Video resolution
- ● Rebuffering (stall) events
- ● Startup delay before playback
- ○ Packet sizes and timing
- ○ IP addresses of the video servers
- ○ Number of parallel connections

2. [2 points]: In Hands-On 1 you looked at a packet capture before doing anything else with it. Which observations should make you suspect a problem with the data? Select all that apply.

- ● A packet with a negative length
- ● A packet far above the maximum packet size
- ● Timestamps that go backwards
- ● Large packets flow mostly server to client
- ○ A handshake before the large packets
- ○ Round-trip times that vary over the trace

3. [3 points]: Attackers are preparing a phishing campaign that imitates the University's login page. You run the campus network and can obtain domain registration data and the logs of the campus DNS resolver. Name **one** data source and **two** features you would use to detect the campaign, ideally before the first email arrives.

(Answer inside the box)

you could use the domain registration data. The features you could use are the similarity of the new registered domain names to the University's login domain, and another could be bursts of related domain registrations within a short period of time and observing that. This can help to show the preparation before the phishing emails are even sen

4. [3 points]: In Hands-On 2, the Log4j scan and the normal web trace were captured at different times of day, so a model with time of day as a feature separates them perfectly. Is time of day a good feature for detecting scans? ○ Yes ● No

Why or why not?

(Answer inside the box)

Time of day is not necessarily helpful for detecting scans. This is what we would call a spurious correlation that is caused by when traces are collected, instead of what actually fundamentally is scanning. Scans can happen at any point in time of day.

Data Acquisition

5. [3 points]: Which of the following are *passive* measurements? Select all that apply.

- ● An ISP collects flow records from its routers
- ● Campus logs every query its DNS resolver answers
- ● Inferring resolution from a capture of your Netflix session
- ○ An app downloads a test file every hour
- ○ A laptop pings a server once a minute
- ○ Measuring latency while a test upload runs

existing user traffic

6. [3 points]: A student reports that Zoom was choppy half an hour ago. You run a speed test from their laptop now and it reports high throughput and low latency. Does this show the network was not the cause? ◯ Yes ● No

Why or why not?

(Answer inside the box)

No, not necessarily. The speed test you run now is measuring the conditions that are now, not 30 min ago when Zoom was choppy. Things like network congestion or jitter could've been occurring in the earlier Zoom session and then are no longer here during the test.

five tuple

7. [3 points]: (a) Name the five header fields that define a flow. In Hands-On 5 it was safe to group packets by only four of them; which one could be left out, and why? (b) Two packets match on all five fields but arrive 45 seconds apart with no packets in between. Would a router using conventional settings put them in the same flow record? Why or why not?

(Answer inside the box)

a) 5-tuple = source IP + destination IP + source port + destination port + transport protocol. Protocol could be omitted when all of the analyzed packets use UDP, since it would be constant then.
b) No, a 45-second gap exceeds the conventional inactivity of ~15-30 seconds, so the first record expires and then the later packet starts a new one. It could have a varied result depending on its configured timeout.

Features and Representation

8. [4 points]: In Hands-On 6 you used `netml` to turn a capture into per-flow STATS features. nPrint instead turns each packet into a fixed-width bitmap of its header bits and lets a model learn which bits matter. Give **one advantage** and **one disadvantage** of *each* approach.

(Answer inside the box)

Netml Stats: Adv: Compact, interpretable summaries help to reduce computation
disadv: When you aggregate you lose packet-level detail and ordering
nprint: Adv: you get to keep the header bits and it lets the model learn useful features automatically
disadv: bitmap doesn't directly capture timing or rates, which must be computed separately.

9. [2 points]: A flow's protocol is TCP, UDP, or ICMP, and you encode it as 0, 1, 2. Which statements are true? Select all that apply.

- ● This is usually fine for a random forest
- ● A linear model treats ICMP as "twice" UDP
- ● One-hot encoding gives three binary columns
- ◯ Min-max scaling fixes it for a linear model
- ◯ Tree models require one-hot inputs
- ◯ The order 0, 1, 2 carries real meaning

10. [3 points]: Some rows in your flow dataset have missing values. Give **one** example from network data where filling in the column average is reasonable, and **one** where it is a bad idea. Explain why for the second.

(Answer inside the box)

Reasonable: When there's just a few randomly missing average packet-length values in otherwise comparable flows
Bad: Protocol codes; averaging across TCP, UDP, and ICMP produces a useless value b/c these are categories not numerical quantities.

Initials: JC

Training and Evaluation

11. [3 points]: You have six months of hourly traffic volumes and want to forecast next month's load to decide whether to add a server. You shuffle all rows randomly, split 80/20, and get low test error. Is that a trustworthy estimate of how well the model will forecast? ◯ Yes ● No

Why or why not?

(Answer inside the box)

Randomly splitting mixes earlier and later observations, and thus nearby hours may be strongly correlated. It doesn't realistically test forecasting future conditions. What you should do is → train on earlier months and evaluate on a later period that's been held-out.

12. [5 points]: A detector labels 1,000 flows; 20 of them are attacks (attack is the positive class). Its confusion matrix:

With attacks positive:
TP = 15 FP = 15
FN = 5 TN = 965

	Predicted attack	Predicted benign
True attack	15	5
True benign	15	965

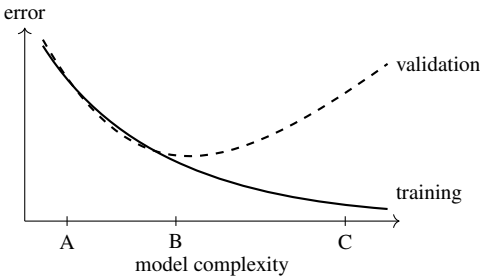
(a) Compute accuracy, precision, and recall. (b) What is the accuracy of a classifier that always says "benign"? What does the comparison tell you, and which curve would you use to compare detectors on this data?

(Answer inside the box)

a) accuracy = $\frac{TP+TN}{1000} = \frac{980}{1000} = 98\%$; Precision = $\frac{TP}{TP+FP} = \frac{15}{30} = 50\%$; Recall: $\frac{TP}{TP+FN} = \frac{15}{20} = 75\%$

b) always predicting benign gives us 98% accuracy; however, it does not detect any attacks. Using accuracy on its own hides performance on the rare positive segment. class imbalance. If we used a precision-recall curve, we could compare detectors on this data.

13. [3 points]: The figure shows training and validation error as model complexity increases.



(a) Which point (A, B, C) has high bias? (b) Which is overfitting? (c) Name one way to reduce overfitting other than choosing a simpler model.

(Answer inside the box)

a) A has high bias. The model is too simple and underfits. High training + validation error
b) C is overfitting. Training error is low while the validation error rises
c) you could add in regularization, which discourages fitting training specific noise

Initials: JC

Assignment 1: Video Quality Inference

14. [3 points]: In Hands-On 3 and Assignment 1 you computed a segment-download-rate feature from encrypted Netflix traffic. (a) How can you find segment boundaries without decrypting anything? (b) Why do segments carry information about resolution?

(Answer inside the box)

a) You can look at the packet timestamps and sizes to group bursts of server-to-client data, thus you can look at between bursts and indicates segment boundaries. You can do this w/out decrypting any traffic. look for bursts of downstream packet traffic seperated by gaps/idle periods

b) Generally, higher resolution video requires more data per second of playing the video. So, by looking at the segment sizes/download rates we can get information about resolution

15. [4 points]: A student's Assignment 1 classifier scores 75% accuracy with a random train/test split, but about 30% when every session from one household is held out for testing. (a) What does this suggest the model learned? (b) An ISP wants to deploy the model in new homes. Which accuracy is the better estimate, and name one change that would help.

(Answer inside the box)

a) The model probably learned household-specific patterns and weird quirks, such as network conditions or devices, which correlate w/ labels but can't generalize to new households.

b) About 30% is the more relevant estimate for deployment in new homes. One change that could help is to collect labeled data from more diverse households and use splits based on household groups for training and validation

Feedback

16. [3 points]: Interest (1=Boring!; 10=Amazing!): 10 Difficulty (1=Too easy; 10=Too hard): 5

One thing you like. One suggestion for improvement:

(Answer inside the box)

I like applying the concepts to real network captures.
Improvement: More real life/ worked examples in class.

Initials: SC